SENSORS DESCRIPTIONS AND THEIR DATA SHEETS

1) HC-SR04 Ultrasonic Sensor

The HC - SR04 (HCSR04 / HC SR04) ultrasonic sensor uses sonar to determine distance to an object like bats or dolphins do. It offers excellent noncontact range detection with high accuracy and stable readings in an easy to use package. Its operation is not affected by sunlight or black material like Sharp rangefinders, (although acoustically soft materials like cloth can be difficult to detect). It comes complete with the ultrasonic transmitter and a receiver module.

Specifications:

- Operating Voltage: +5V
- Theoretical Measuring Distance: 2cm to 400cm
- Practical Measuring Distance: 2cm to 80cm
- Accuracy: 3mm
- Measuring angle covered: <15°
- Operating Current: <15mA
- Operating Frequency: 40Hz

2) Water Flow Sensor YF-S201 Flowmeter G1/2 1–30 L/min

This sensor " YF - S201 Flowmeter / Flow meter " sits in line with your water line and contains a pinwheel sensor to measure how much liquid has moved through it. There's an integrated magnetic hall effect sensor that outputs an electrical pulse with every revolution. The hall effect sensor is sealed from the water pipe and allows the sensor to stay safe and dry.
The sensor comes with three wires: red (5-24VDC power), black (ground) and yellow (Hall effect pulse output). By counting the pulses from the output of the sensor, you can easily calculate water flow. Each pulse is approximately 2.25 milliliters. Note this isn't a precision sensor, and the pulse rate does vary a bit depending on the flow rate, fluid pressure and sensor orientation. It will need careful calibration if better than 10% precision is required. However, its great for basic measurement tasks.

The pulse signal is a simple square wave so its quite easy to log and convert into liters per minute using the following formula.

Pulse frequency (Hz) / 7.5 = flow rate in L/min.

Specifications:

- Model: YF-S201
- Sensor Type: Hall effect
- Working Voltage: 5 to 18V DC (min tested working voltage 4.5V)
- Max current draw: 15mA @ 5V
- Output Type: 5V TTL
- Working Flow Rate: 1 to 30 Liters/Minute
- Working Temperature range: -25 to +80℃
- Working Humidity Range: 35%-80% RH
- Accuracy: ±10%
- Maximum water pressure: 2.0 MPa
- Output duty cycle: 50% +-10%
- Output rise time: 0.04us
- Output fall time: 0.18us

- Flow rate pulse characteristics: Frequency (Hz) = 7.5 * Flow rate (L/min)
- Pulses per Liter: 450
- Durability: minimum 300,000 cycles
- Cable length: 15cm
- 1/2" nominal pipe connections, 0.78" outer diameter, 1/2" of thread
- Size: 2.5" x 1.4" x 1.4"

3) HX711 Weight Pressure Sensor Module

HX711 (HX 711) Load Cell Amplifier Module uses 24 high-precision ADC converter chip hx711, is designed for high-precision electronic scale and design, with two analog input channels. The input multiplexer selects either Channel A or B differential input to the low-noise programmable gain amplifier (PGA). Channel A can be programmed with a gain of 128 or 64, corresponding to a full-scale differential input voltage of ±20mV or ±40mV respectively, when a 5V supply is connected to AVDD analogue power supply pin. Channel B has a fixed gain of 32. .The input circuit can be configured to provide a bridge voltage electrical bridge (such as pressure, weight) sensor model is an ideal high-precision, low-cost sampling front-end module.There is no programming needed for the internal registers. All controls to the HX711 are through the pins.

Specifications:

- Two selectable differential input channels
- On-chip active low noise PGA with selectable gain of 32, 64 and 128
- On-chip power supply regulator for load-cell and ADC analog power supply
- On-chip oscillator requiring no external component with optional external crystal
- On-chip power-on reset
- Simple digital control and serial interface: pin-driven controls, no programming needed
- Simultaneous 50 and 60Hz supply rejection
- Current consumption including on-chip analog power supply regulator: normal operation < 1.5mA, power down < 1uA
- Operation supply voltage range: 2.6V ~ 5.5V
- Operation temperature range: -40 ~ +85C

4) GY-BMP280 Atmospheric Pressure Sensor Module

The GY - BMP280 (GY BMP280 / BMP 280) Barometer Sensor is a breakout board for Bosch BMP280 high-precision and low-power digital barometer. It can be used to measure temperature and atmospheric pressure accurately. It can be connected to a microcontroller with I2C. BMP280 is an absolute barometric pressure sensor especially designed for mobile applications. Its small dimensions and its low power consumption allow for the implementation in battery powered devices such as mobile phones, GPS modules or watches.

As its predecessor BMP180, BMP280 is based on Bosch’s proven Piezo-resistive pressure sensor technology featuring high accuracy and linearity as well as long term stability and high EMC robustness. Numerous device operation options offer highest flexibility to optimize the device regarding power consumption, resolution and filter performance. A tested set of default settings for example use case is provided to the developer in order to make design-in as easy as possible.

Specifications:

- Model: GY-BMP280-3.3
- Chip: BMP280
- Power supply: 3V/3.3V DC
- Peak current: 1.12mA
- Air pressure range : 300-1100hPa (equi. to +9000…-500m above sea level)
- Temperature range: -40 … +85 °C

- Digital interfaces: I²C (up to 3.4 MHz) and SPI (3 and 4 wire, up to 10 MHz)
- Current consumption of sensor BMP280: 2.7µA @ 1 Hz sampling rate

DHT11 Temperature and Humidity Sensor Module

The DHT 11 is commonly used as a temperature and humidity sensor. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers.

The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of ±1°C and ±1%.

Specifications:

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Output: Serial data
- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Resolution: Temperature and Humidity both are 16-bit
- Accuracy: ±1°C and ±1%

6) 5-Channel Line Tracking Sensor Module

5 Channel Line Tracking Sensor Module (BFD-1000)
BFD – 1000 specifically designed as a black line (white), especially suitable for the complex of the black and white line, cross black and white line detection, it has 6 roads high sensitivity infrared sensor (3 escapes patrol, 1 road), to the recognition of black and white line accurately.

General Specifications:
✅5 channel high sensitivity sensor
✅High accuracy for tracking black line.
✅Distance sensor at the front, distance can be adjusted
✅Special touch sensor design, making the robot design simple
✅Digital output signal
✅Indicator LEDs

Technical Specifications:
✅Detection distance: 0 – 4 cm (black and white line sensors), 0 – 5 cm (adjustable distance detection)
✅Input voltage: 3.0 V – 5.5 V